

Convergent mediation | SAP Billing and Revenue Innovation Management

SAP Convergent Mediation by DigitalRoute and SAP Convergent Invoicing

Proving the capacity to scale in high-volume data transfers

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The data transfer performance between the SAP Convergent Mediation solution by DigitalRoute and the SAP Convergent Invoicing package underwent rigorous tests. The result? Transfer rates reached up to 640 million records per hour and demonstrated scalability along with smooth data collection, cleansing, and processing.

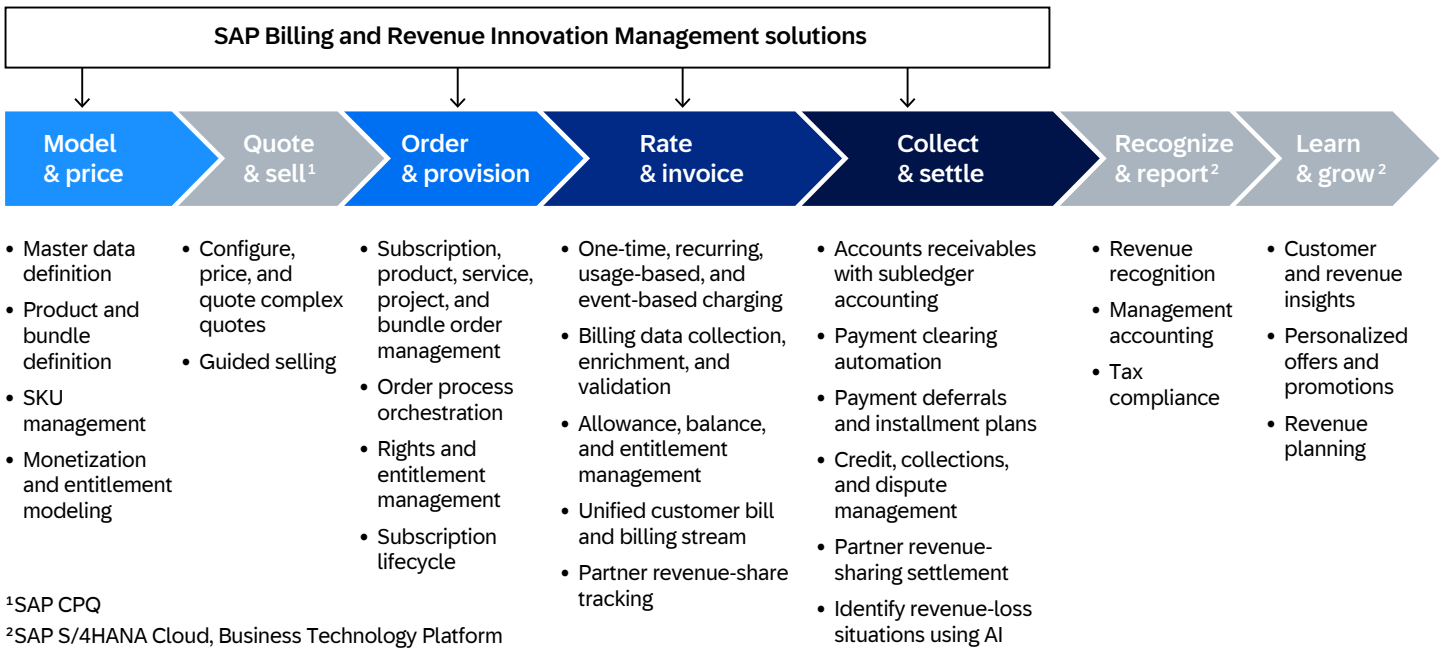
Streamlining the complexity of billing and revenue management

As companies move faster and pivot more often to keep up with customer demand, outdated data transfer processes can hold them back—stifling growth and innovation. Managing ever-growing volumes of business and customer transactions with rigid systems is no longer sustainable. To truly enable business growth, your core system must be as agile and scalable as the products and services you bring to market.

SAP Billing and Revenue Innovation Management solutions provide the capabilities and tools businesses need to keep up with changing expectations (see Figure 1). They support B2B and B2C customer transaction management at scale and with increasing ease—from mediation, rating, billing, and invoicing of digital subscriptions and pay-per-use services to multi-sided revenue sharing with partners and complex settlement processes.

To prove the capacity of SAP Billing and Revenue Innovation Management to manage growth and complexity SAP conducted a series of performance tests on the solutions' various components. These tests demonstrated high-performance scalability, which is critical for minimizing revenue leakage, delivering accurate billing for customers, and settling partner revenue-share payouts smoothly as businesses grow.

Figure 1: Transaction management coverage of SAP Billing and Revenue Innovation Management solutions



[1]SAP CPQ
[2]SAP S/4HANA Cloud, Business Technology Platform

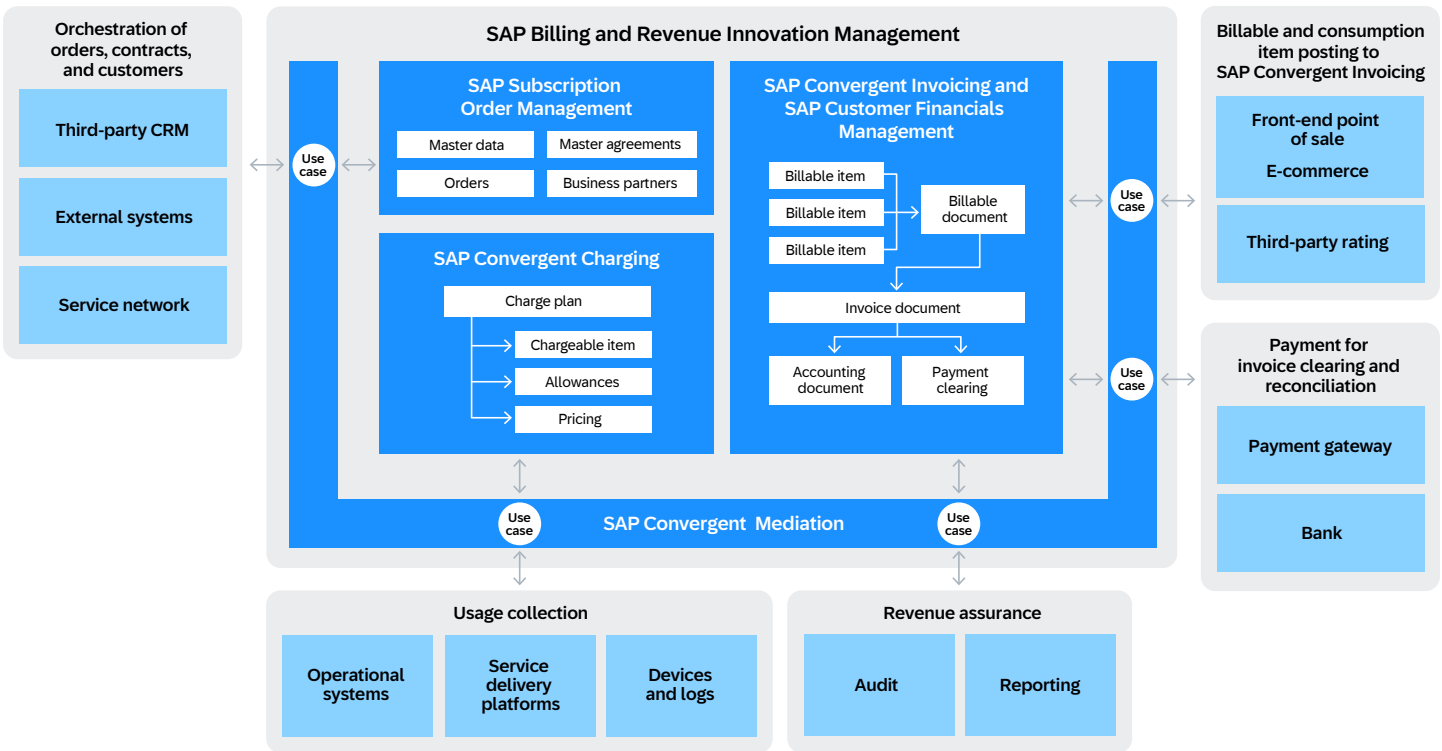
Enabling scalable capture and consumption of transactions

One of the key components of SAP Billing and Revenue Innovation Management is the SAP Convergent Mediation solution by DigitalRoute. This solution serves as the standard integration layer. It automatically captures and cleanses high volumes of raw transactions from various sources and front-end channels and feeds them into the SAP Convergent Invoicing package.

This process helps ensure data quality, enriches transactions, and provides high-scale loading capabilities for final transactions. As a result, data from sales activities across multiple commerce platforms can be integrated into a single accounting system—empowering finance teams to track financial data confidently.

There are various scenarios in which SAP Convergent Mediation interacts with SAP Convergent Invoicing, including pricing, invoicing, accounting, and payments (see Figure 2). Among these, one of the most prominent scenarios for handling high volumes is billable item posting into the SAP Convergent Invoicing package and the SAP Customer Financial Management solution. Here, the solution collects, converts, and processes raw transactions into billable items (BITs) or consumption items (CITs) by executing preconfigured workflows.

Figure 2: Common use cases for SAP Convergent Mediation by DigitalRoute



The processing steps that SAP Convergent Mediation by DigitalRoute workflows can perform include:

Mapping of external transaction formats to BIT format

SAP Convergent Mediation offers a flexible mapping layer to convert external protocols and data formats —such as CSV, XML, JSON, and Protobufs—into the BIT format for SAP Convergent Invoicing.

Master data enrichment

For proper categorization or grouping, BIT records need to be enriched with external metadata, which can be done through database lookups or external REST or other API calls.

Record splitting

When containing multiple billable transactions, an incoming transaction is split into several BITs to allow for separate grouping, aggregation, and billing.

Aggregation

Multiple partial or related transactions can be correlated and aggregated into a single BIT.

Deduplication

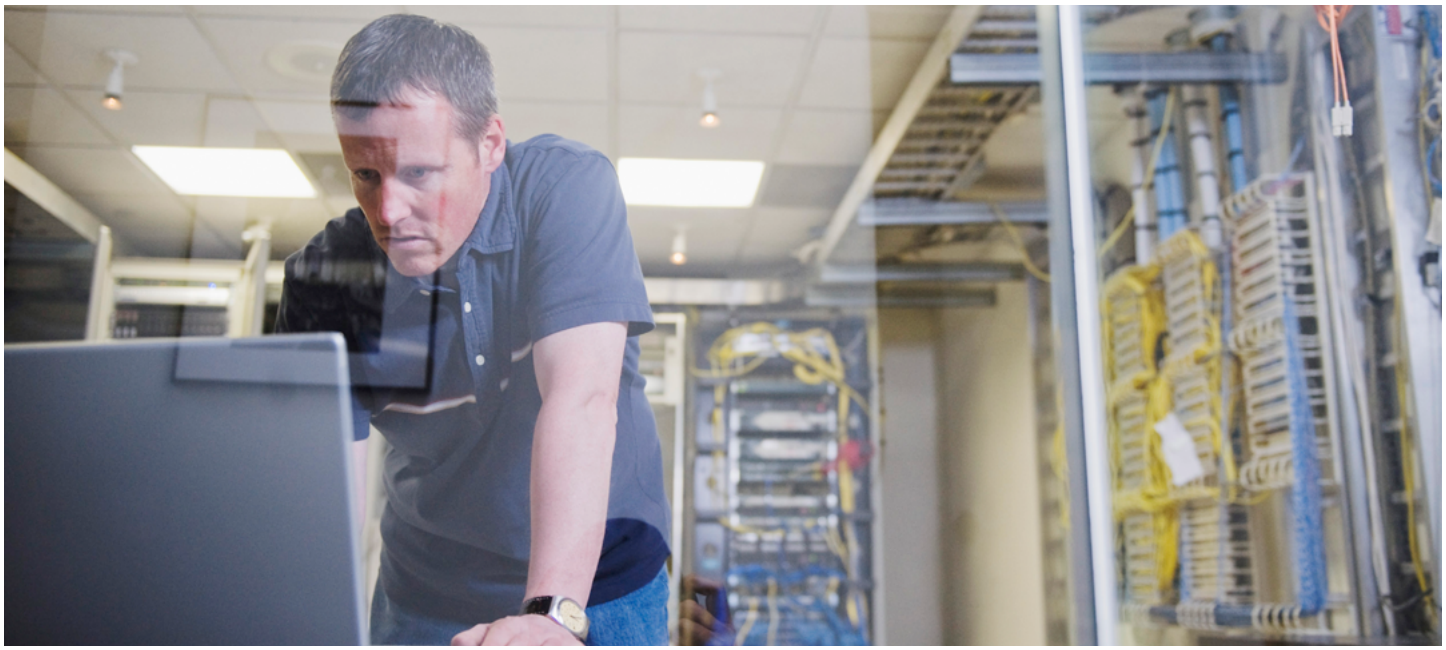
Duplicate transactions can be removed before loading BITs into SAP Convergent Invoicing using a function that is highly scalable and capable of deduplicating billions of historical and current transactions.

Scalable creation of BITs

A comprehensive, high-performance approach to BIT creation is offered, with the ability to parallel load BITs into SAP Convergent Invoicing.

Uploading of unpriced transactions

In addition to BITs, consumption items (CITs) can be uploaded into SAP Convergent Invoicing to be priced at a later stage. The structure of CITs and BITs is similar, so the performance metrics in this document apply to both.





Setting the performance test scope for scalability

The integration between SAP Convergent Mediation and SAP Convergent Invoicing underwent a high-volume load test to demonstrate data transfer rates capable of processing hundreds of millions of records within 60 minutes. This level of scalability indicates the potential to handle even higher volumes, if needed.

The result? Data transfer rates reaching up to 640 million events and records every hour, showcasing near-perfect scalability.

This outcome is particularly valuable for companies looking to shift their business models from a product-based approach to a subscription basis or from indirect to direct sales. These new business models significantly increase the number of consumer accounts to track—from thousands to hundreds of millions, even billions of transactions. In fact, the number of purchases can easily exceed the number of accounts by tenfold, highlighting the need to process extremely high data volumes efficiently.

Setting the performance test scope for SAP Convergent Mediation

The scope of this test focused on the direct transfer of billable items from SAP Convergent Mediation to SAP Convergent Invoicing while bypassing SAP Convergent Charging (see Figure 3). This scenario is ideal when complex pricing logic isn't necessary or when prices are managed by other systems, such as web shops.

Notably, the Java connector provides a development library that enables Java applications to communicate with on-premise SAP solutions through a remote function call (RFC) protocol. Doing so eliminated the need for additional components.

The primary goal of this test was to demonstrate high-volume data transfer capabilities and substantial throughput between the two systems. We also aimed to validate scalability and determine the optimal bulk size for transfers. Sending data in excessively small batches could lead to high CPU consumption, while overly large batches might result in memory issues.

Figure 3: The test scope of the integration of SAP Convergent Mediation and SAP Convergent Invoicing

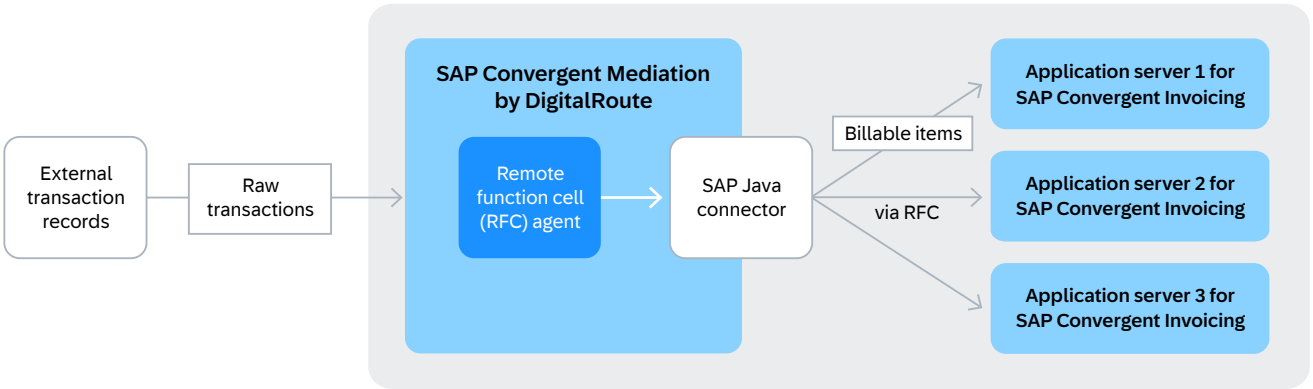
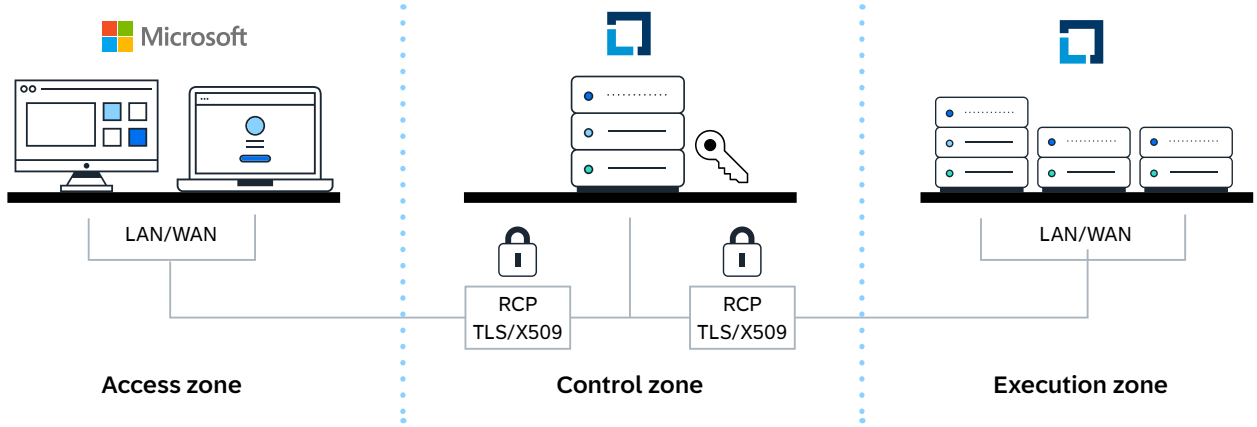


Figure 4: The three-layer architecture of SAP Convergent Mediation by DigitalRoute



The setup

The performance test was executed using the 4.2 version of SAP Convergent Mediation with the SAP HANA database. The solution has a cluster runtime architecture with three layers as illustrated in Figure 4:

Access zone

The user interfaces on desktops and web-based interfaces are made available.

Control zone

Platform processes are connected to SAP HANA to coordinate and communicate events with other processes. The platform process is also used in design time, where workflow objects for SAP Convergent Mediation are developed, configured, activated, and distributed to registered execution context processes.

Execution zone

This layer includes execution contexts and the working container for actual runtime loads, where workflows, batches, and tasks are completed. It usually consists of multiple execution context processes, running on multiple hosts.

In our performance test, one host was dedicated to the control zone, and three hosts were dedicated to the execution zone. Each execution zone host performs the execution context processes.

BITs were generated from SAP Convergent Mediation and bulk loaded into SAP Convergent Invoicing using the RFC protocol. To achieve this, multiple workflows in SAP Convergent Mediation were created.

The batch workflow that triggers the RFC protocol reads a CSV file containing input parameters for the performance test, the bulk size of the BITs upload, and the number of RFC requests for sending the BITs (see Figure 5). Based on this input, the data that should be in the BITs is created and sent via the workflow bridge agent to the next workflow that sends the actual RFC request.

Figure 5: Batch workflow for triggering the RFC protocol



The real-time workflow receives the data from the batch workflow and maps it into the correct BIT format (see Figure 6). The BITs are then sent to SAP Convergent Invoicing over the RFC protocol. The number of BITs per RFC request—or bulk size—is 750. SAP Convergent Mediation inspects the RFC function response and creates a summary record of each RFC call, which is sent to the inter-workflow.

Figure 6: Real-time workflow for dispatching the RFC protocol

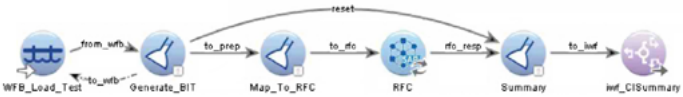
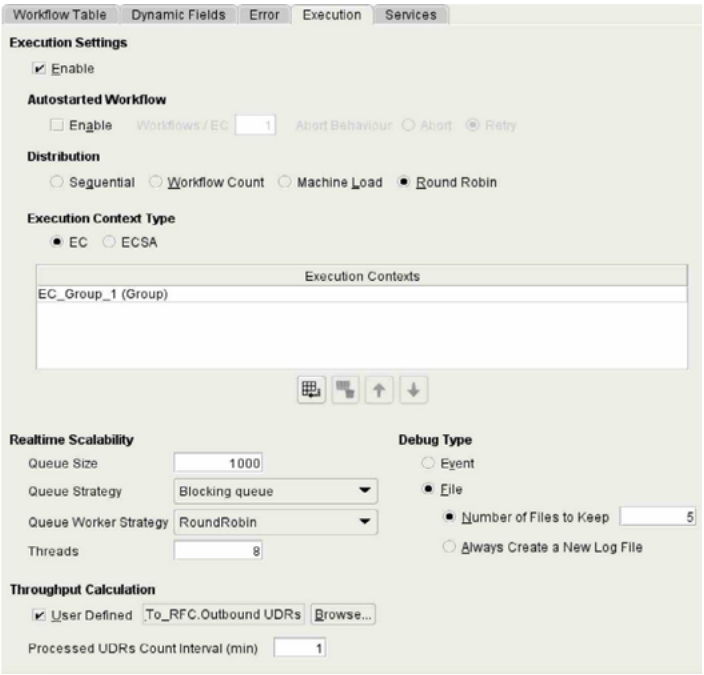
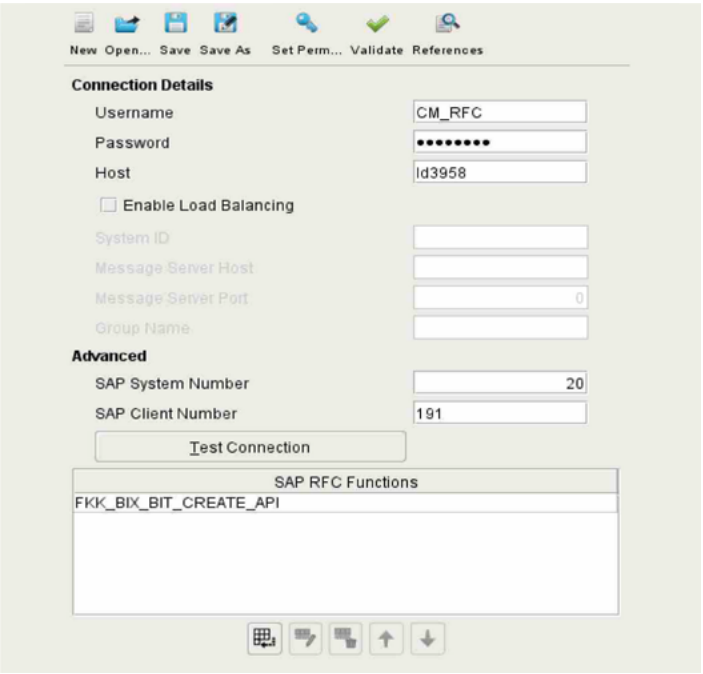


Figure 7: Configuration of the workflow (left) and RFC profile (right)





The workflow is set to use eight threads and a queue size of 1,000. At the same time, the RFC profile is programmed with the RFC function FKK_BIX_BIT_CREATE_API. In this case, load balancing is not configured by default. See Figure 7 for configurations.

Additionally, the RFC agent is configured according to the RFC profile, and its settings are defaulted (see Figure 8).

The batch workflow creates a CSV file with output summary record for each RFC call executed by real-time workflow – dispatch RFC (See Figure 9). This confirms that all RFCs have been received by SAP Convergent Invoicing.

Figure 8: RFC agent settings

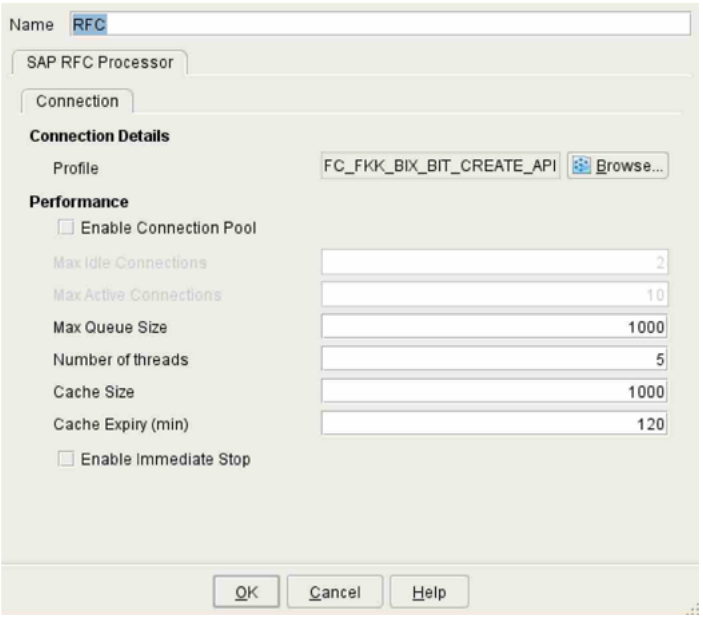


Figure 9: Batch workflow for SAP Convergent Invoicing





Test setup of the hardware environment

SAP Convergent Mediation

SAP Convergent Mediation by DigitalRoute was installed on five identical hosts based on Intel architecture. The following characteristics apply to one host:

CPU type:	Intel(R) Xeon(R) CPU E5-2630 v3 at 2.40GHz
Sockets:	2
Cores (total):	16 (8 for each socket)
Threads (total):	32 (2 for each core)
Memory:	64 GB

SAP Convergent Invoicing

The environment for SAP Convergent Invoicing consisted of different host types based on Intel architecture:

Host type A:	9 hosts used
CPU type:	Intel(R) Xeon(R) CPU E5-2630 v3 at 2.40GHz
Sockets:	2
Cores (total):	16 (8 for each socket)
Threads (total):	32 (2 for each core)
Memory:	64 GB

Host type B:	3 hosts used
CPU type:	Intel(R) Xeon(R) CPU E5-2680 v4 at 2.40GHz
Sockets:	2
Cores (total):	28 (14 for each socket)
Threads (total):	56 (2 for each core)
Memory:	64 GB

Test setup of SAP Convergent Invoicing

SAP Convergent Invoicing was accompanied by six servers with the SAP HANA database and 25 servers with the ABAP version of the SAP NetWeaver application. Fifteen of the application servers were located on a single server, while the remaining ten shared one physical server. No virtualization technology was used.

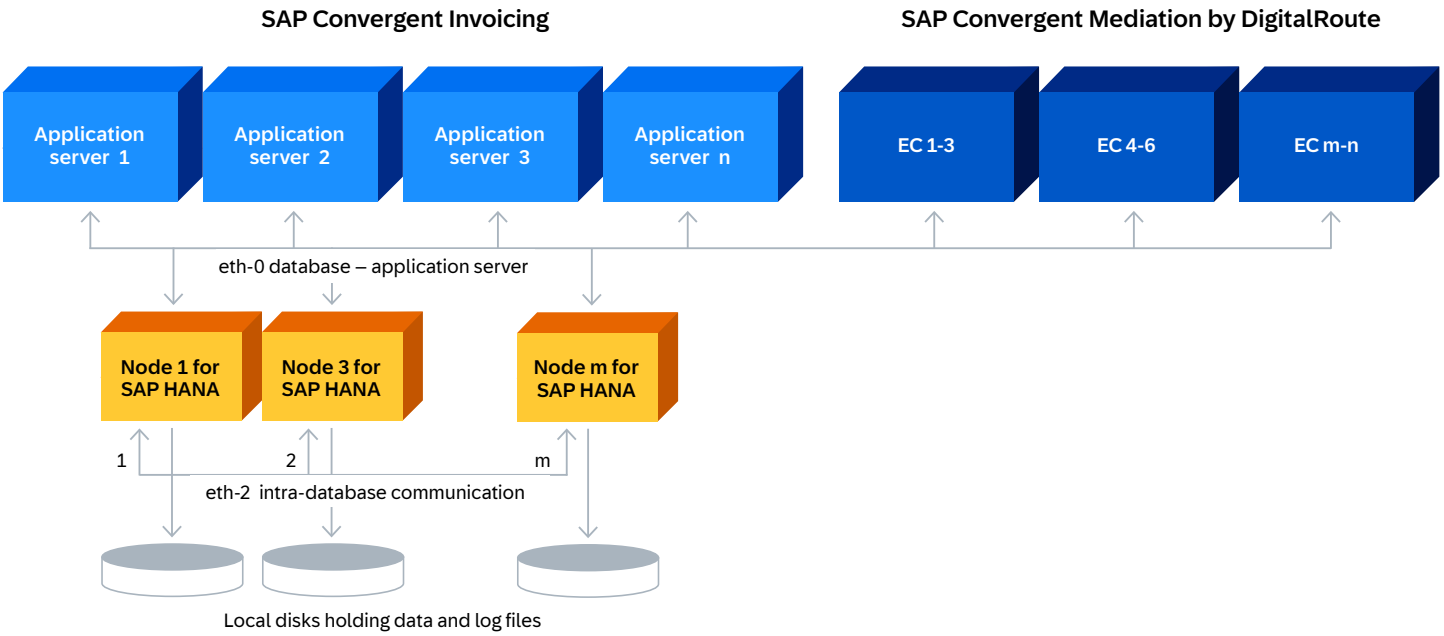
The general setup of the environment is illustrated in Figure 10. On the right side are servers supporting SAP Convergent Mediation by DigitalRoute, with three execution contexts for each host. Data from this system is sent to the hosts for SAP Convergent Invoicing, using the same network that connects them to the database hosts. Since load distribution is managed through a Java connector and log-on group, each execution context can send data to any application server on the side of SAP Convergent Invoicing.

The SAP HANA version used was 2.00.052.00, while SAP Convergent Invoicing was on the 2009 version of SAP S/4HANA.

SAP Convergent Mediation by DigitalRoute connects to SAP Convergent Invoicing through the Java connector adapter and performs RFCs to send the billable items directly to SAP Convergent Invoicing.

For the test, the number of dialog work processes handling the RFCs was initially set to 120 for each server; however, it later became clear this high count was unnecessary. More than 40 processes were rarely needed. Nevertheless, this high number was chosen for the test to help ensure that the number of processes did not create a bottleneck.

Figure 10: Topology of the environment supporting SAP Convergent Invoicing with SAP HANA

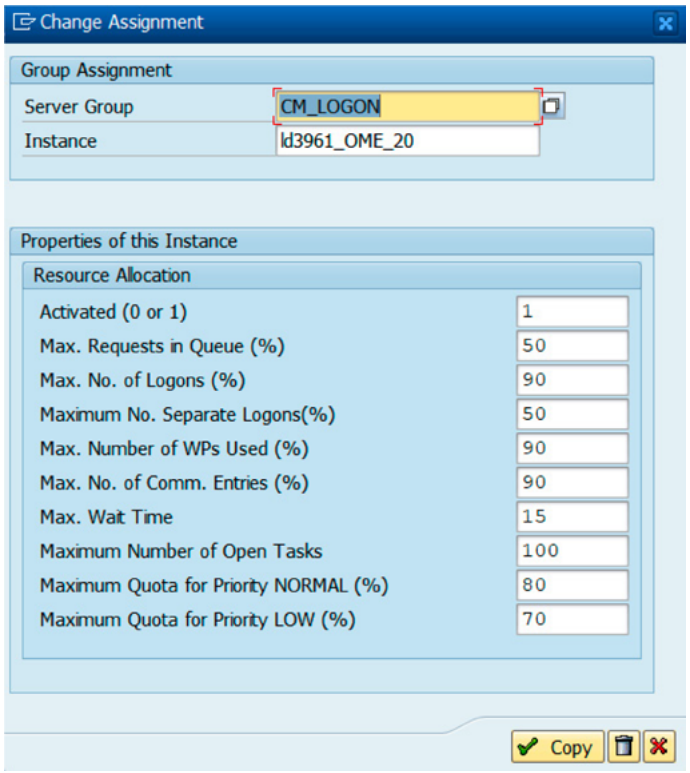


The parameter rdisp/rfc_max_queue was increased from 5% to 50% to allow a higher volume of RFC calls in the dispatcher queue (see Figure 11). Other parameters controlling resource consumption on the application servers remained at their standard values.

A dedicated log-on group was also created to control which servers would receive requests. Depending on the test, the number of servers assigned to the logon group was adjusted to maintain a one-to-three ratio between the servers supporting SAP Convergent Mediation and those supporting SAP Convergent Invoicing. Pretests indicated that this ratio served as a good rule of thumb for performance testing when both systems used the same type of hosts.

Additional tests were conducted with different workload distribution algorithms provided by the logon group. It was found that circular workload distribution achieved the highest overall throughput. The alternative “best-performance” algorithm tended to switch servers only if the current server was already experiencing slight overload. However, the differences between the approaches were minimal.

Figure 11: The resource allocation parameters for the test



Detailing the results

The scope of the test was to prove the capability to transfer high volumes of data between SAP Convergent Mediation and SAP Convergent Invoicing. During the test, a throughput of more than 640 million BITs each hour was achieved with four servers supporting SAP Convergent Mediation feeding 12 servers with SAP Convergent Invoicing.

The second objective was to validate the scalability of the integration. In this case, an excellent result was also achieved, with almost optimal linear scalability (see Figure 12).

All tests were run for a minimum of 20 minutes to demonstrate the long-term stability of the data transfer. This is reflected in the CPU usage graphs from the different servers (see figures 13–16).

In the figures on the next page, lines in orange represent the usage of the servers with SAP Convergent Invoicing while those in blue reflect the usage of servers supporting SAP Convergent Mediation. The ratio between SAP Convergent Mediation and SAP Convergent Invoicing hosts was consistently maintained at 1 to 3.

Figure 12: Optimal linear scalability of the integration between SAP Convergent Mediation and SAP Convergent Invoicing

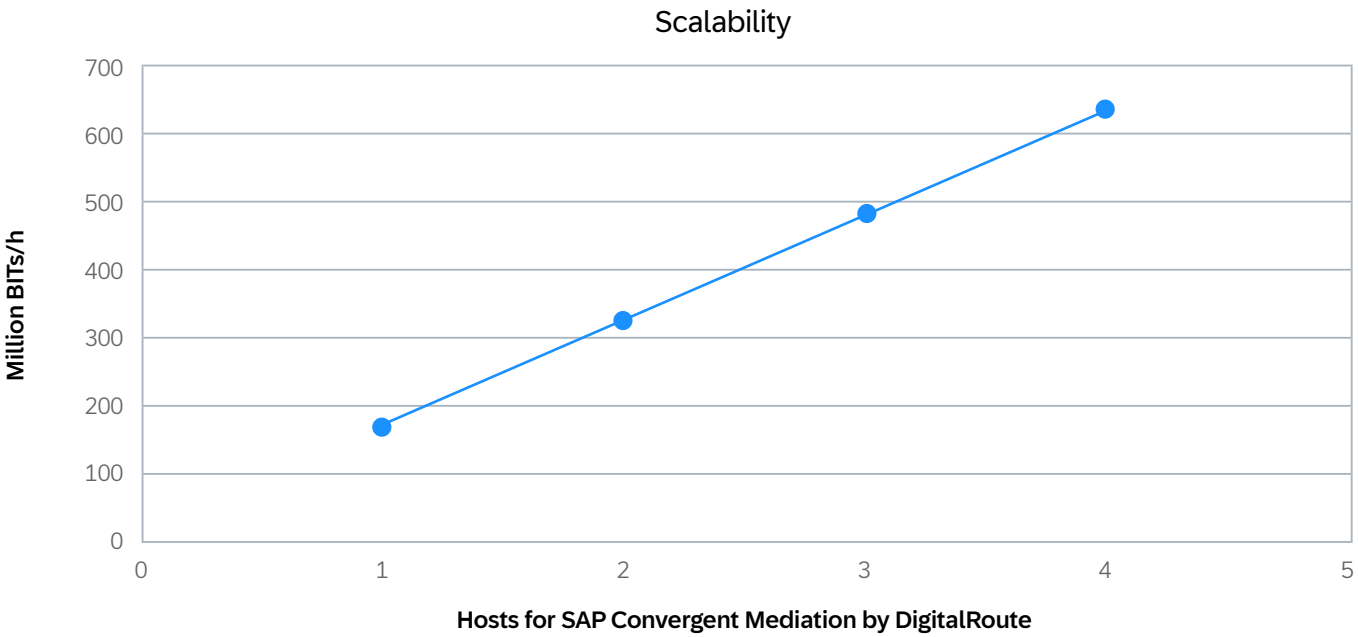


Figure 13: CPU utilization of one host with SAP Convergent Mediation by DigitalRoute and three hosts with SAP Convergent Invoicing

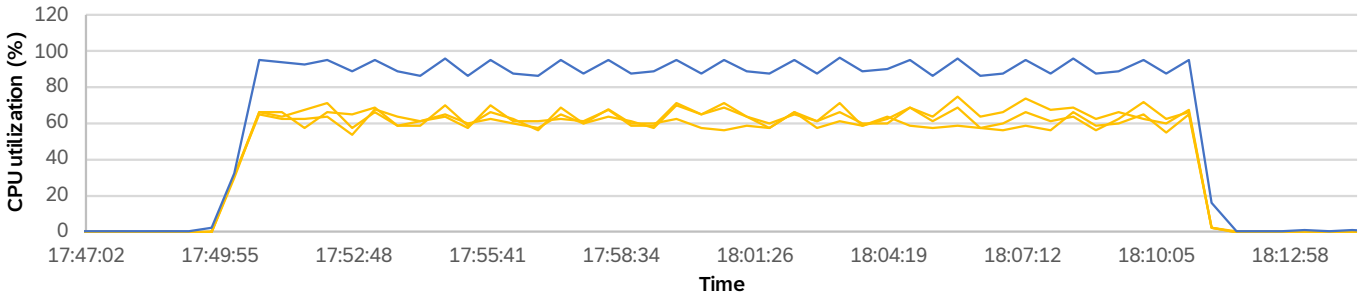


Figure 14: CPU utilization of two hosts with SAP Convergent Mediation by DigitalRoute and six hosts with SAP Convergent Invoicing

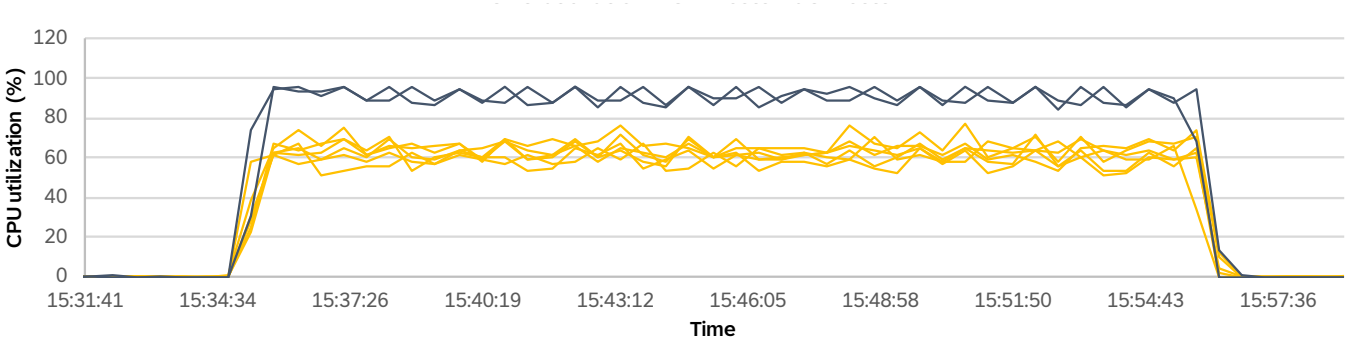


Figure 15: CPU utilization of three hosts with SAP Convergent Mediation by DigitalRoute and nine hosts with SAP Convergent Invoicing

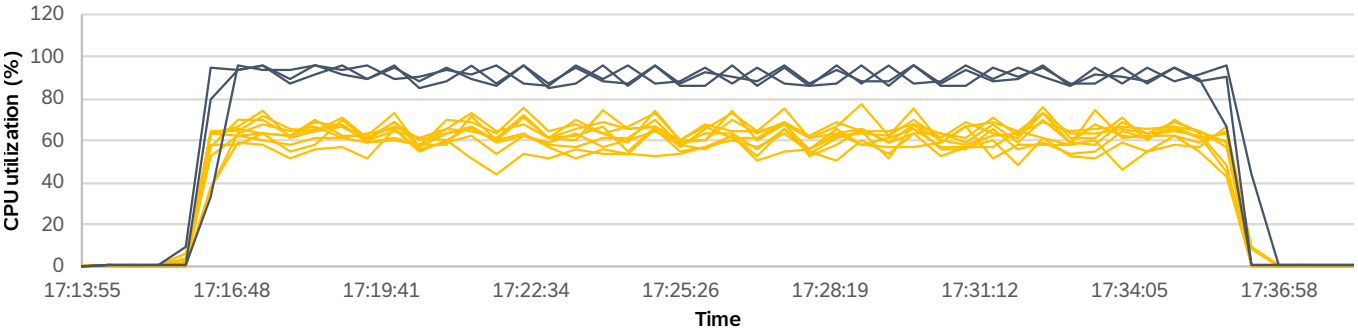
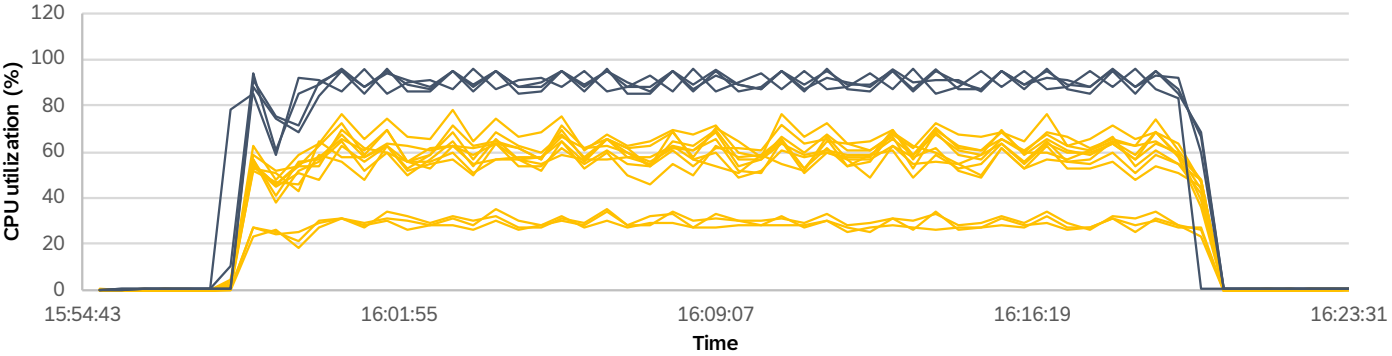


Figure 16: CPU utilization of four hosts with SAP Convergent Mediation by DigitalRoute and 12 hosts with SAP Convergent Invoicing



In the last set of tests, two distinct levels of CPU usage were observed on the servers with SAP Convergent Invoicing. This variation is due to three of the 12 servers being of a different type, with 56 logical CPUs, while the remaining nine servers had only 32. Since incoming RFC calls were distributed in a circular fashion, the larger servers received a relatively lighter workload and lower CPU usage. The resulting server input and output confirm this behavior, as no difference in input and output was observed between the 12 hosts for SAP Convergent Invoicing (see figures 17–20).

Figure 17: Network input and output when using one host for SAP Convergent Mediation and three hosts for SAP Convergent Invoicing

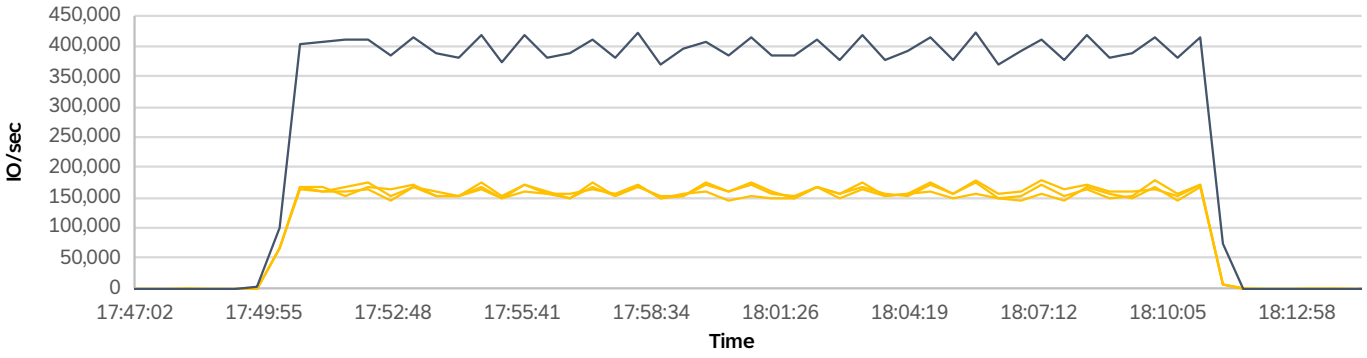
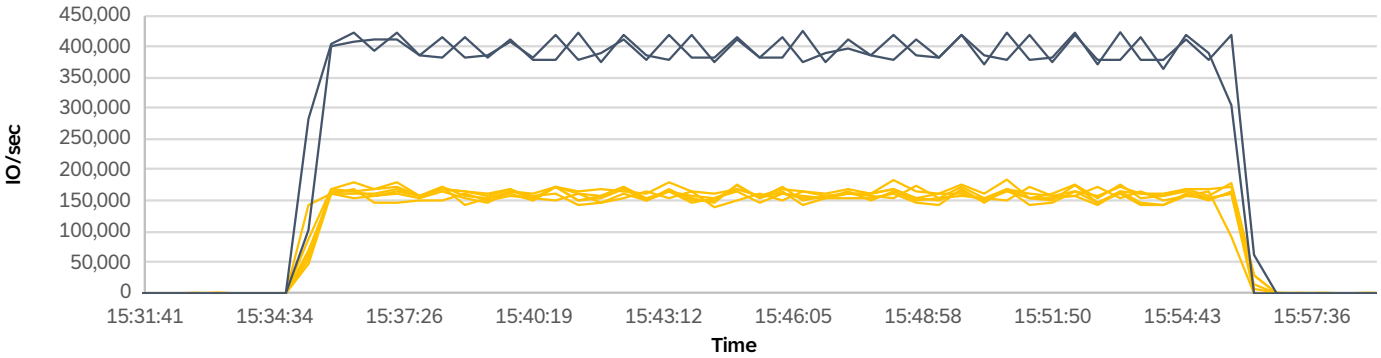


Figure 18: Network input and output when using two hosts for SAP Convergent Mediation and six hosts for SAP Convergent Invoicing



Based on our initial tests, the network traffic on the hosts for SAP Convergent Mediation could be expected to be three times higher than the network traffic measured on the hosts for SAP Convergent Invoicing. However, as shown in figures 13–16, this is not the case. The hosts for SAP Convergent Invoicing forward the data to SAP HANA after verification, generating additional traffic.

Figure 19: Network input and output when using three hosts for SAP Convergent Mediation and nine hosts for SAP Convergent Invoicing

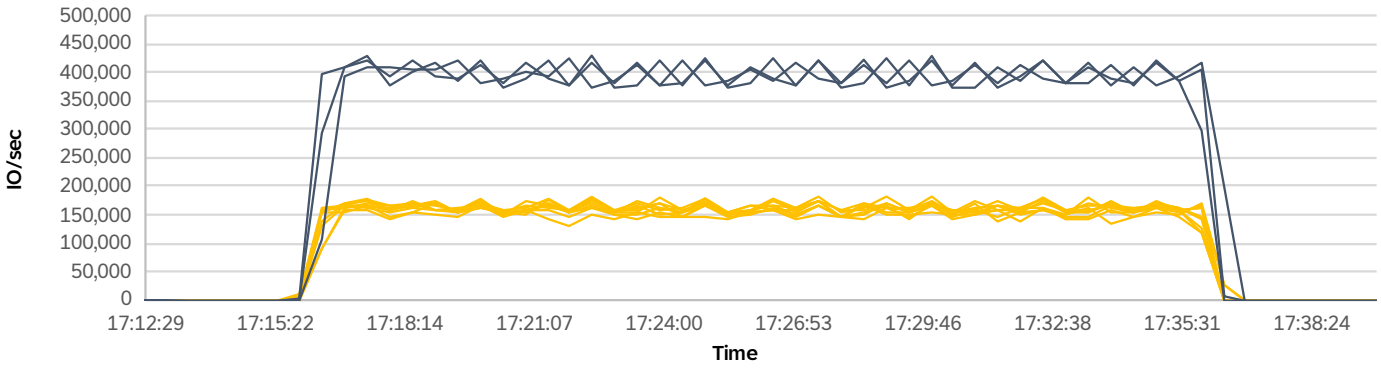
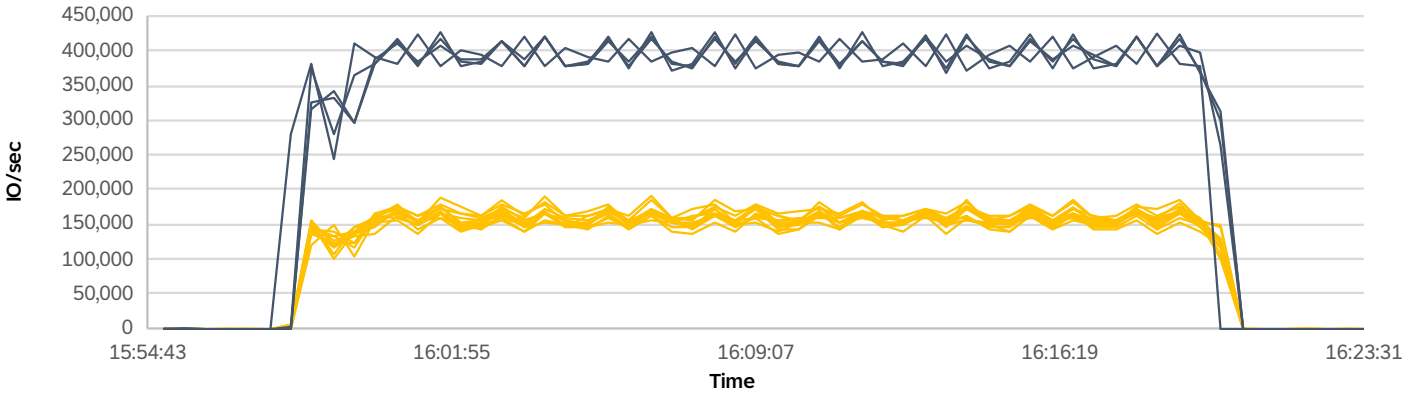


Figure 20: Network input and output when using four hosts for SAP Convergent Mediation and 12 hosts for SAP Convergent Invoicing



Learn more

To find out more about SAP Convergent Mediation and SAP Billing and Revenue Innovation Management, please contact your local SAP representative or visit sap.com/brim.

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